ECEN721: Optical Interconnects Circuits and Systems
Spring 2026

Lecture 9: Mach-Zehnder Modulator Transmitters

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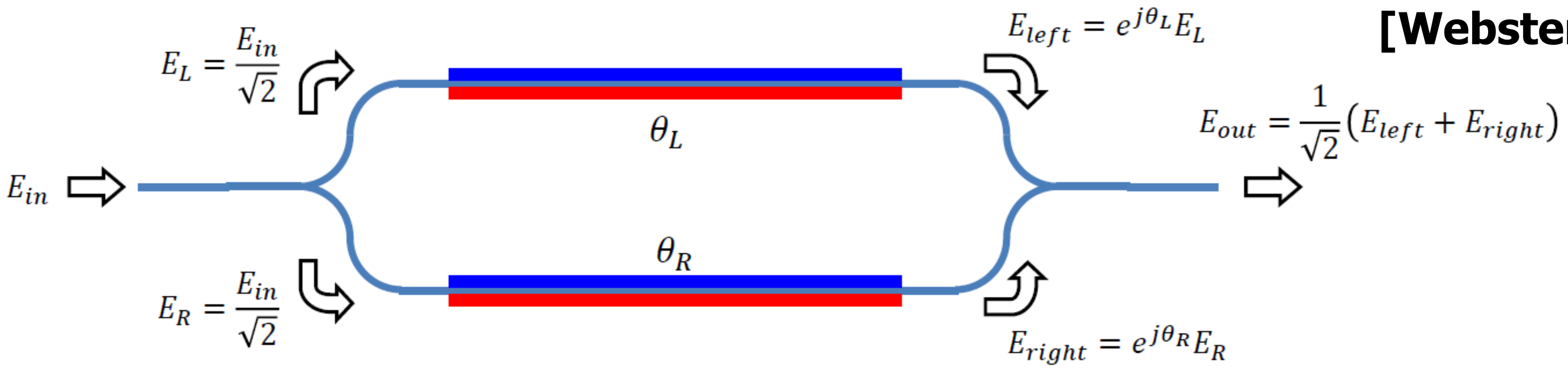
Announcements

- Exam 1 Mar 3
  - In class
  - One double-sided 8.5x11 notes page allowed
  - Bring your calculator
  - Covers through Lecture 6
- Homework 3 due Mar 19
- Reading
  - Sackinger Chapter 8

Mach-Zehnder Modulator (MZM)

[Webster CSICS 2015]



- An optical interferometer is formed with the incoming light split, experiencing phase shifts through the two paths, and then recombined
- Assuming no loss and a perfect 50/50 splitter/combiner

$$\Delta\phi = \frac{(\theta_R - \theta_L)}{2} \qquad \phi = \frac{(\theta_R + \theta_L)}{2}$$

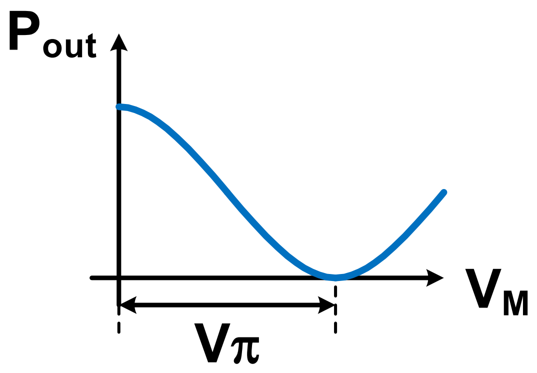
Field Response

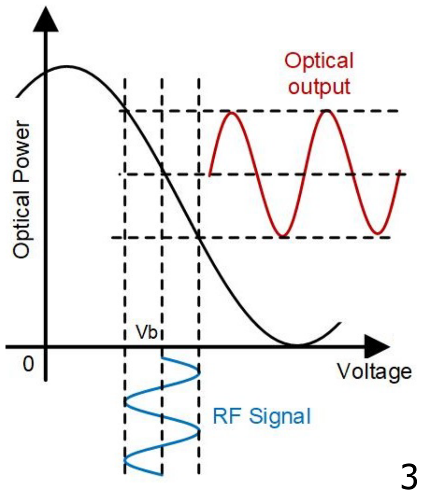
$$E_{out} = E_{in}\cos(\Delta\phi)e^{j\phi}$$

Intensity Response

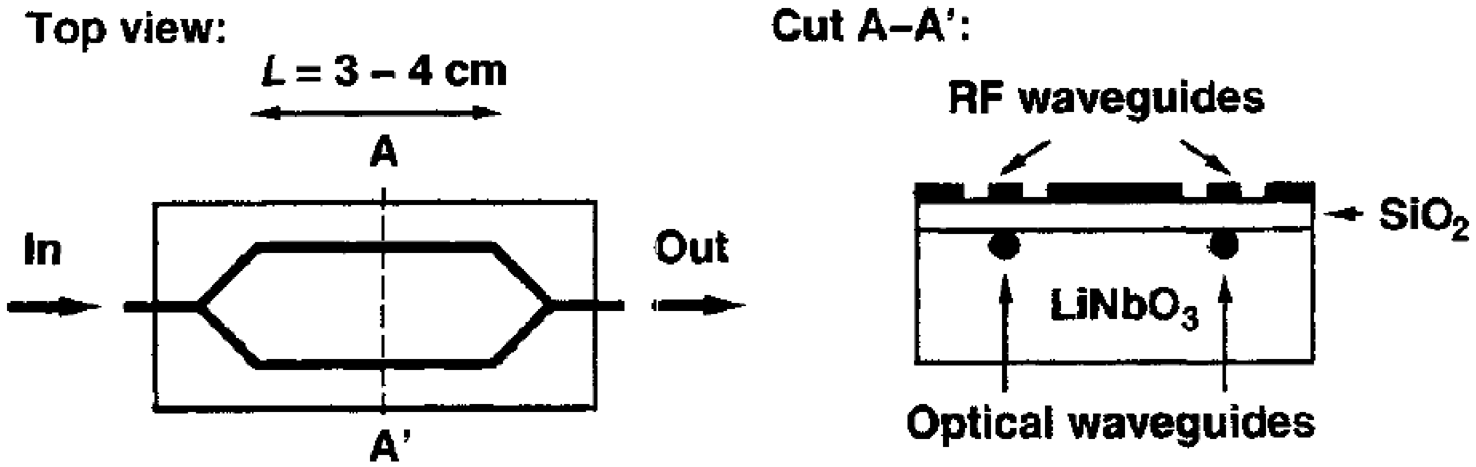
$$P_{out} = |E_{out}|^2 = \frac{1}{2}|E_{in}|^2[1+\cos(\theta_R - \theta_L)]$$

$$\boxed{\frac{P_{out}}{P_{in}} = \frac{1}{2}[1+\cos(\theta_R - \theta_L)] = \frac{1}{2}\left[1+\cos\left(\pi\cdot\frac{V_M}{V\pi}\right)\right]}$$



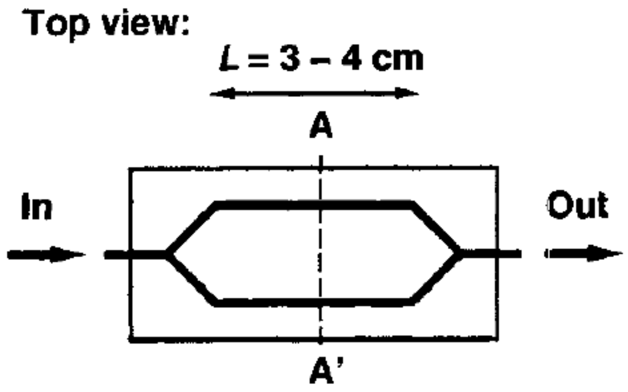


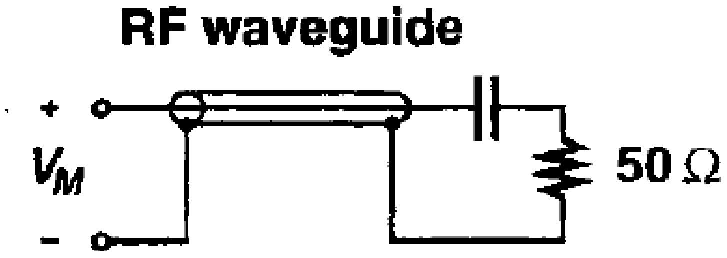
Single or Dual-Drive



- Single-Drive MZM
  - Only one are is driven in a single-ended manner
  - While only requiring a single high-speed input signal, there is generally some chirp in the output signal
  - Need to apply the full V_{Π} to one are to get maximum extinction ratio
- Dual-Drive MZM
  - Both arms are driven in a differential/push-pull manner
  - This ideally results in no chirp at the output
  - Only need to apply $\pm V_{\Pi}/2$ on the two arms to get maximum extinction ratio

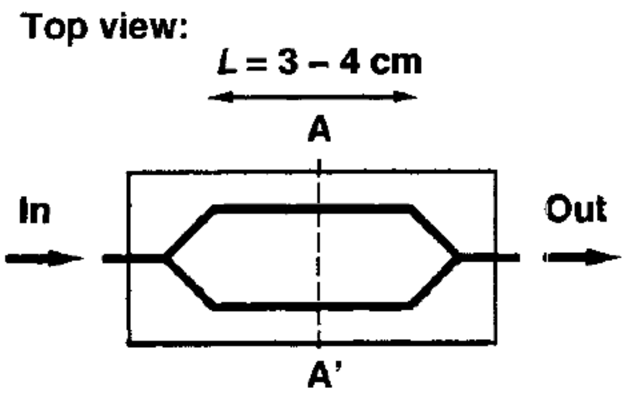
V_Π*L Product





- The amount of phase shift generated by an MZM is proportional to voltage applied and the length of the phase shifter
- Thus, an MZM figure of merit is the V_Π*L product
- Typical values
  - Lithium Niobate: 14Vcm
  - Silicon (Depletion-Mode): 4Vcm
  - Silicon (MOS Capacitor): 0.2Vcm
- These large V_Π*L products lead to long controlled-impedance electrodes which must be terminated
- A key challenge is matching the propagation speed of the electrical modulation signal with the optical beam

Chirp Parameter



$$\alpha = \frac{v_{M1}^{pp} + v_{M2}^{pp}}{v_{M1}^{pp} - v_{M2}^{pp}}$$

where v_{M1}^{pp} and v_{M2}^{pp} are the voltage swings on the two modulator arms.

With differential signaling $v_{M1}^{pp} = -v_{M2}^{pp}$ and the chirp is ideally zero.

If $v_{M1}^{pp} < -v_{M2}^{pp}$ then we can actually have negative chirp and potential pulse compression when passed through a fiber with positive *D*

If $v_{M1}^{pp} = v_{M2}^{pp}$, then we get a purely phase modulated signal and the MZM can be used for phase modulation (QPSK), rather than amplitude modulation.

Silicon Free Carrier Plasma Dispersion Effect

- The refractive index of silicon can be changed through the free-carrier plasma dispersion effect where the electron and hole densities change the refractive index
- Unfortunately, this also changes the waveguide's absorption (loss)
- This effect is utilized for all present high-speed silicon photonic modulators

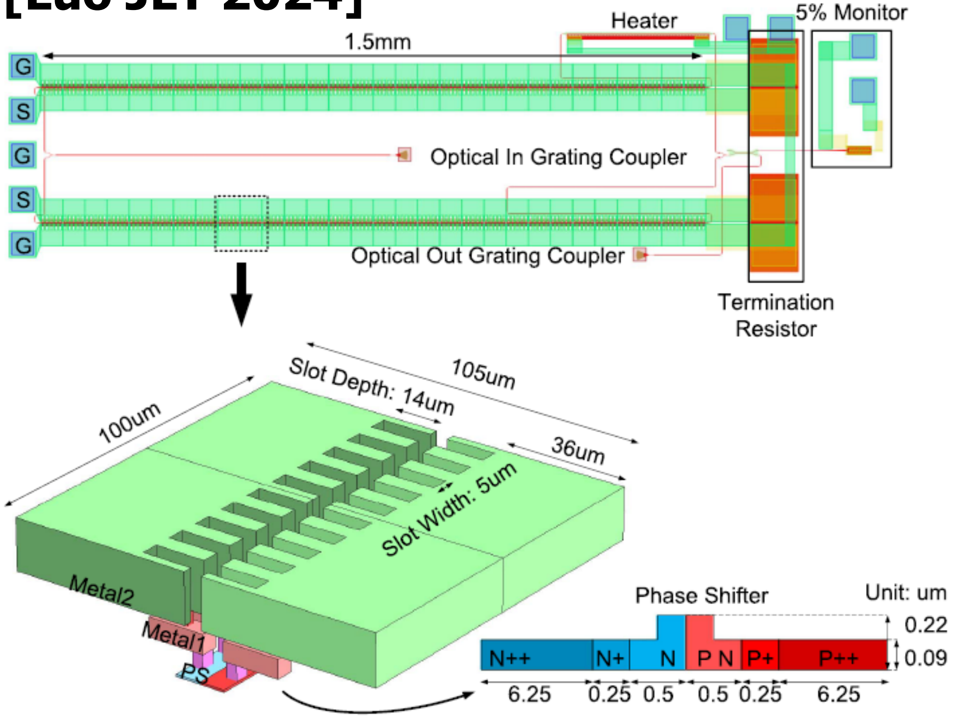
$$\left.\begin{aligned} \Delta n &= \Delta n_e + \Delta n_h = -8.8 \times 10^{-22}(\Delta N_e) - 8.5 \times 10^{-18}(\Delta N_h)^{0.8} \\ \Delta \alpha &= \Delta \alpha_e + \Delta \alpha_h = 8.5 \times 10^{-18}(\Delta N_e) + 6.0 \times 10^{-18}(\Delta N_h) \end{aligned}\right\} \lambda\text{=1550nm}$$

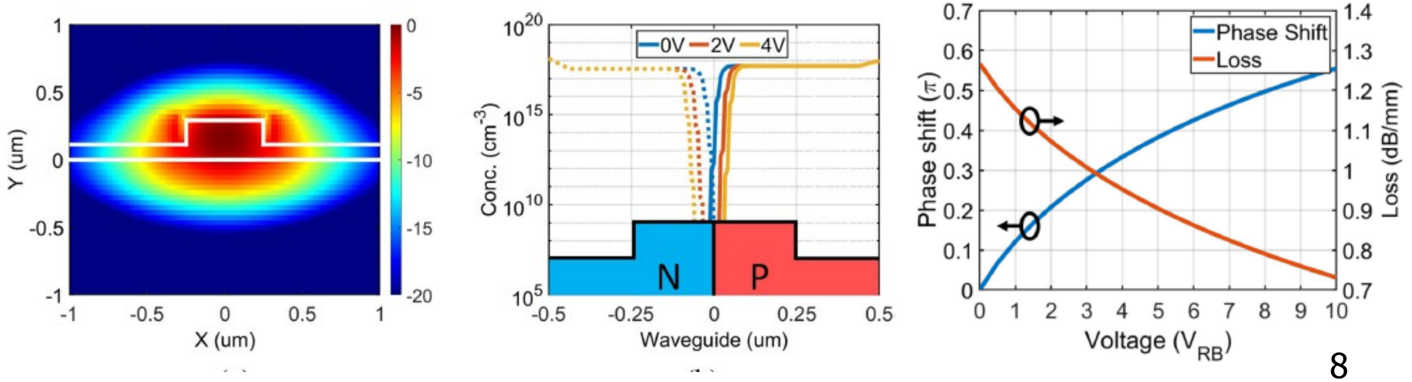
$$\left.\begin{aligned} \Delta n &= \Delta n_e + \Delta n_h = -6.2 \times 10^{-22}(\Delta N_e) - 6.0 \times 10^{-18}(\Delta N_h)^{0.8} \\ \Delta \alpha &= \Delta \alpha_e + \Delta \alpha_h = 6.0 \times 10^{-18}(\Delta N_e) + 4.0 \times 10^{-18}(\Delta N_h) \end{aligned}\right\} \lambda\text{=1310nm}$$

Silicon Photonic MZMs

- Silicon waveguide is doped as a PN junction with the depletion region modulated as a function of the applied reverse bias voltage
- The resultant change in carrier density within the depletion region causes the refractive index to change through the plasma dispersion effect
- Tradeoffs in V_{Π}, bandwidth, and loss
- Typical values
  - $V_{\Pi}L$ = 1.5-2V*cm
  - V_{Π} = 6-7V
  - BW = 30-35GHz
- Results in mm-scale devices typically driven in a traveling-wave manner

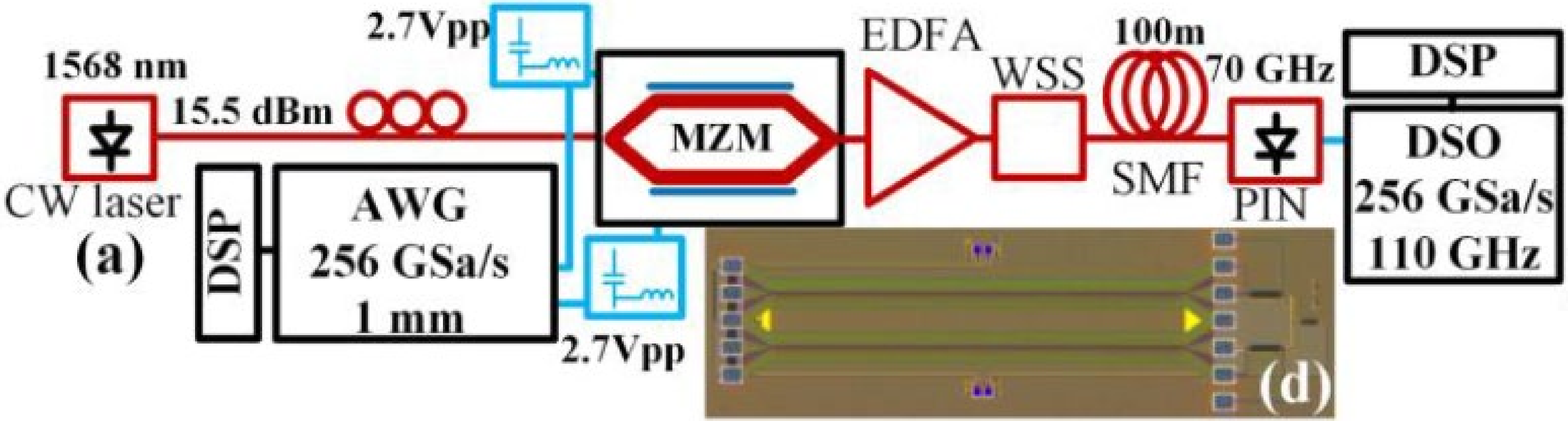
[Luo JLT 2024]

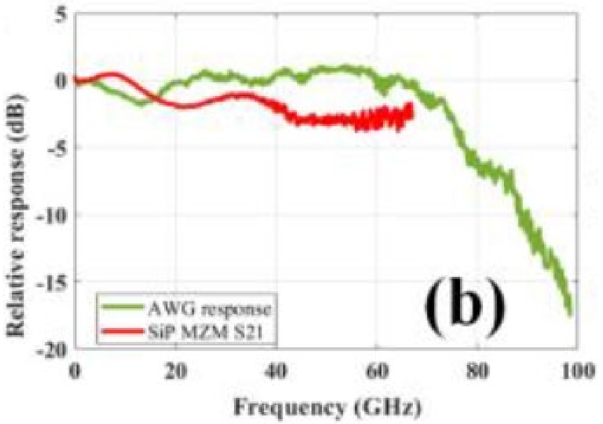




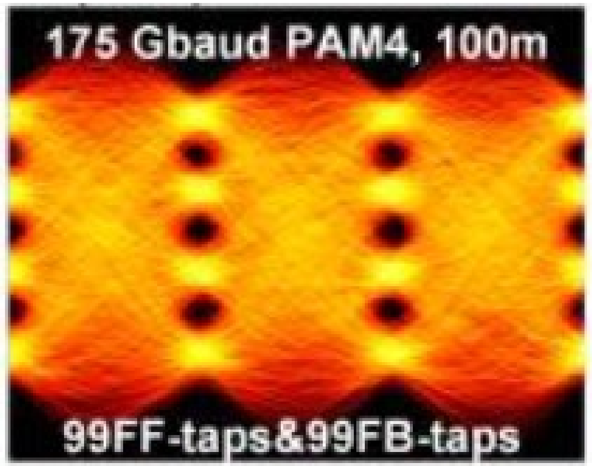
42GHz Silicon Photonic MZM

[Ostrovskis OFC 2025]





350Gb/s PAM4

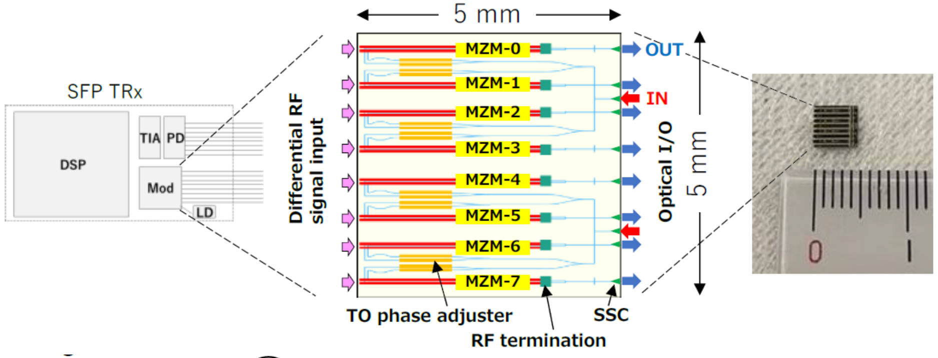


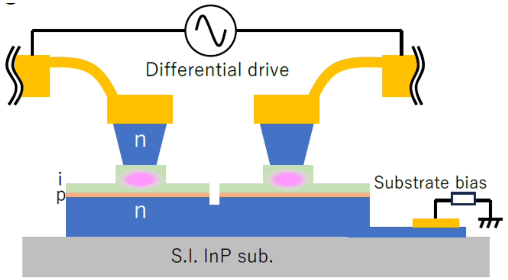
- 42GHz bandwidth with 2.5V bias
- $V_{\Pi}L$ = 1.7V*cm
- L=1.5mm
- 3.6dB insertion loss plus 2.5dB per grating coupler
- 350Gb/s PAM4 operation achieved with scope (RX) equalization of 99-tap FFE and 99-tap DFE

InP MZMs

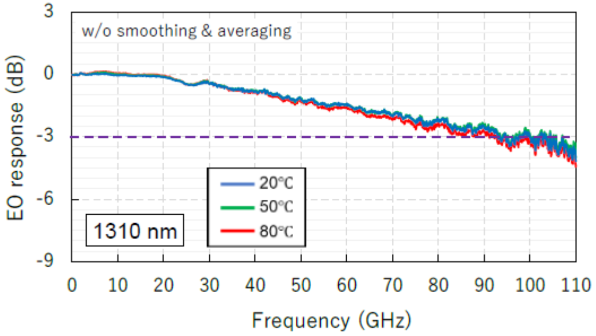
- InP MZMs are commonly used for long-reach transmission due to small V_{Π}, high bandwidth, and low chirp
- Compatable with differential drive
- InP is generally an expensive material
- State of the art is 100GHz bandwidth that achieved 430Gb/s PAM6
  - 65Ω differential impedance
  - Mismatch with 100Ω AWG required a 4001 tap T/2-spaced FFE
  - RX DSP used 61-tap T/2-spaced FFE

[Ogiso OFC 2025]
8-Channel Module

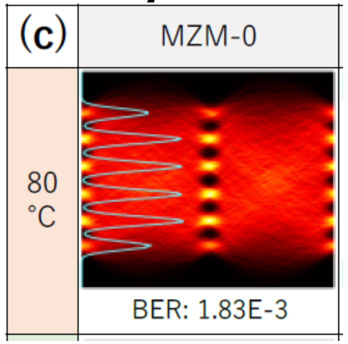




100GHz Bandwidth

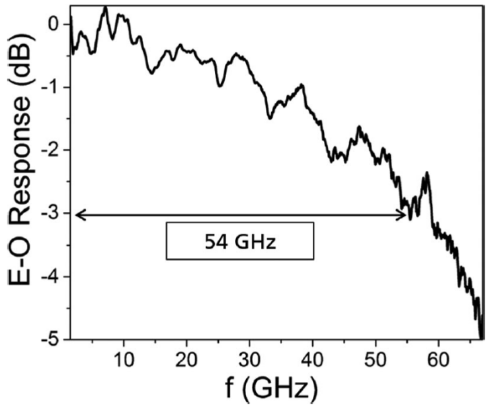


430Gb/s PAM6

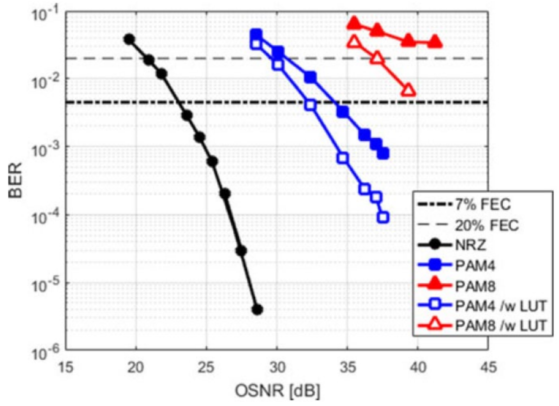


InP Monolithic DFB Laser + MZM

[Lange JLT 2018]

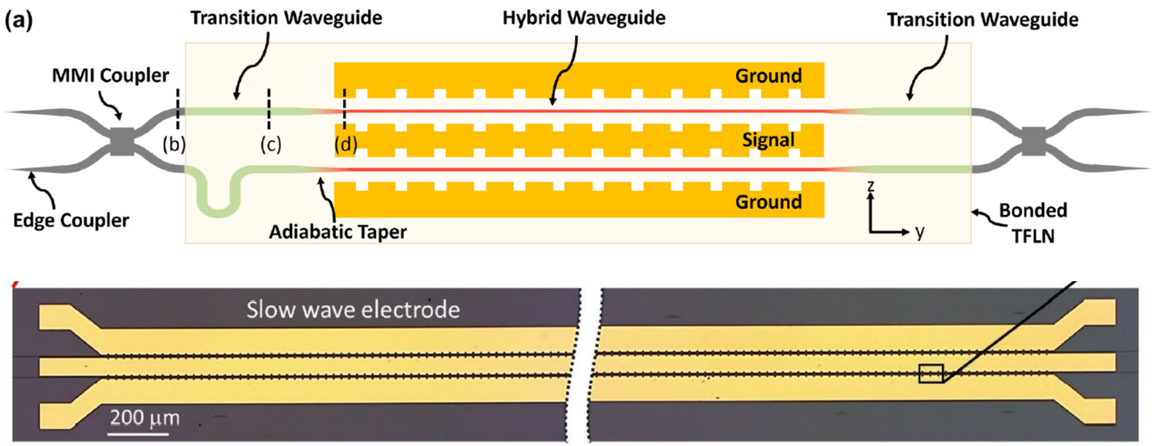


100GBd BER

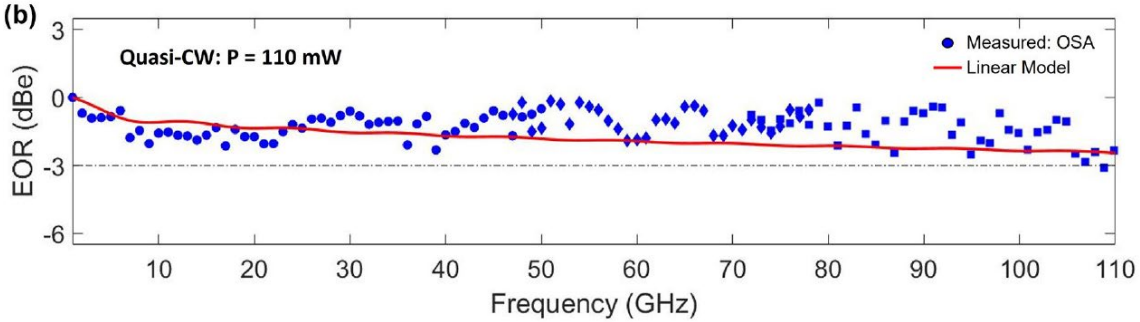


- InP platform allows for monolithic DFB laser and MZM integration
- MZM parameters
  - 54GHz bandwidth
  - $V_{\Pi} = 2V$
  - $V_{\Pi}L = 0.55V*cm$
- 200Gb/s PAM4 operation achieved with 7% FEC
- 300Gb/s PAM8 possible with 20% FEC and LUT-based equalizer

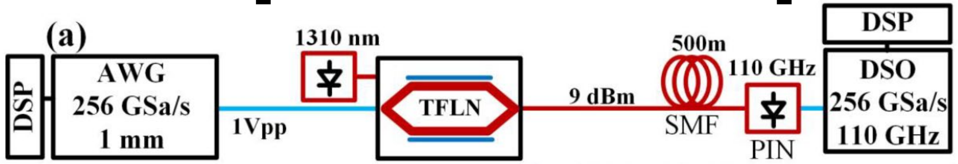
Thin-Film Lithium Niobate (TFLN) MZM



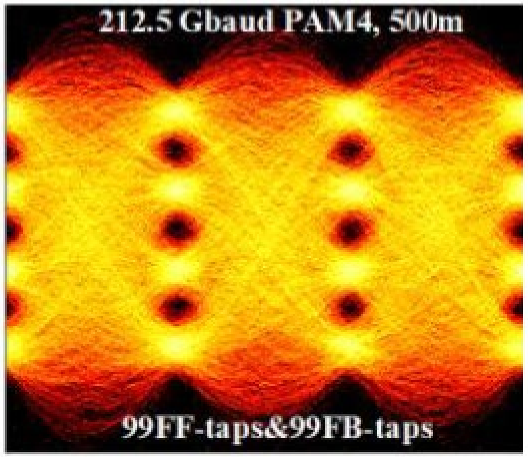
[Valdez SciRep 2022]



[Ostrovskis OFC 2025]



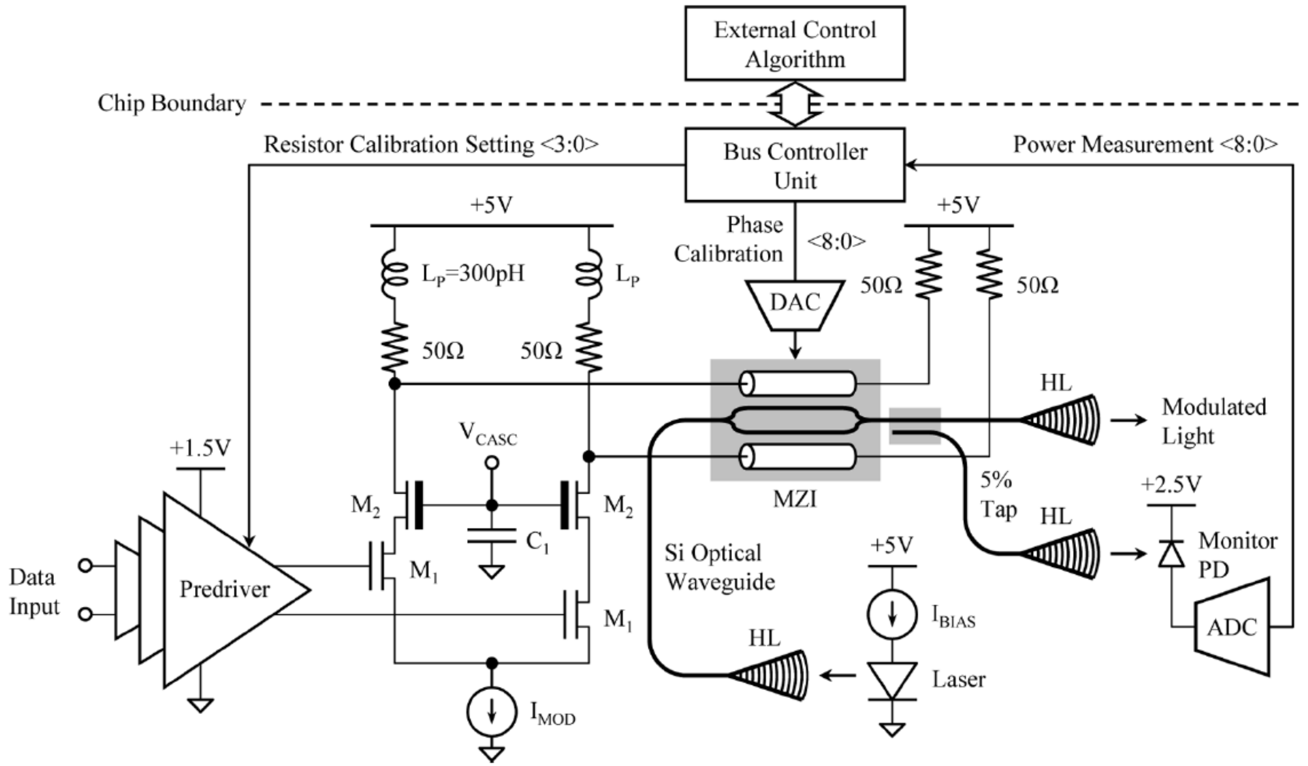
425Gb/s PAM4

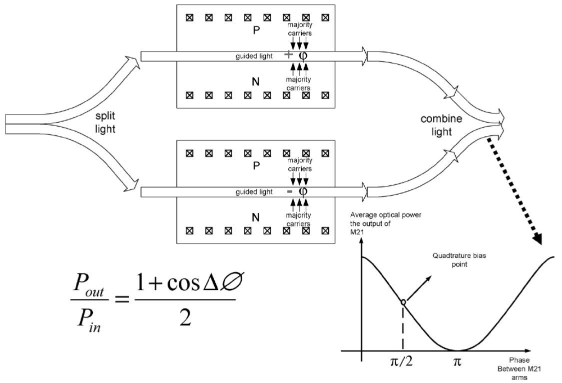


- Low loss and high bandwidth allow for longer modulators with low V_{Π} values
- 110GHz MZM realized by bonding LN on Si waveguides
  - Potential to leverage existing SiP platforms
  - $V_{\Pi}L$ = 3.1V*cm
  - L=5mm
  - 1.8dB insertion loss
- 425Gb/s PAM4 demonstrated with similar device

Traveling-Wave MZM Driver

[Analui JSSC 2006]

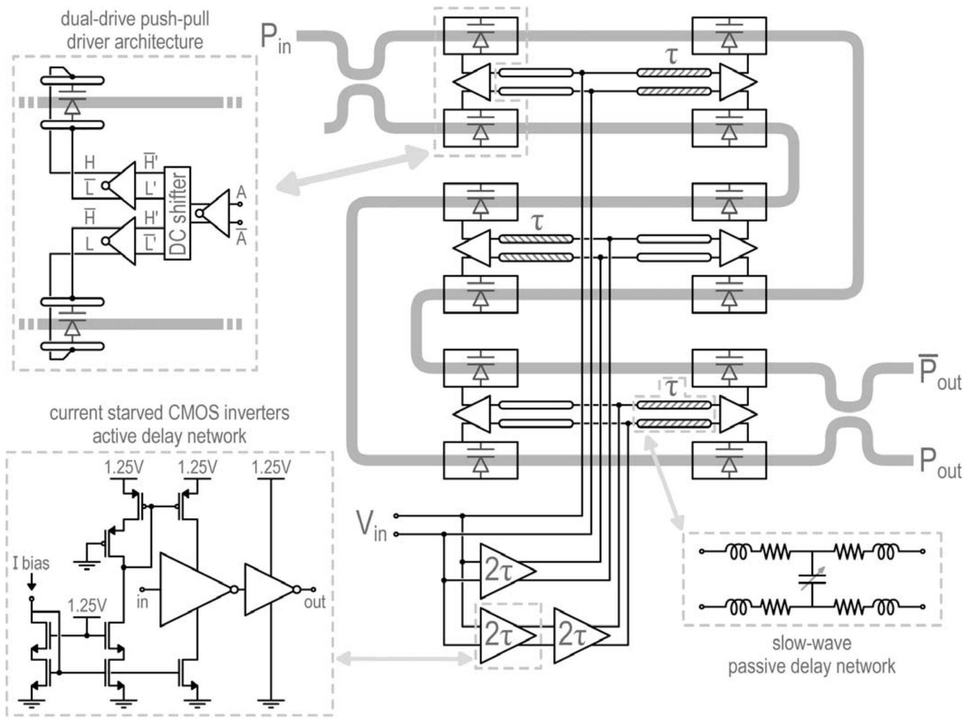


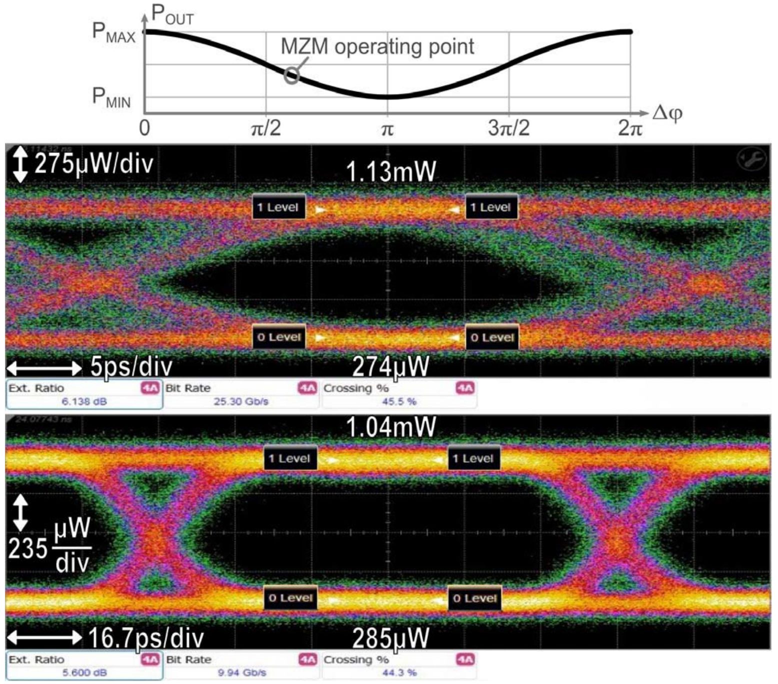


- Depletion-mode MZM is driven with a $5V_{ppd}$ signal

Distributed MZM Driver

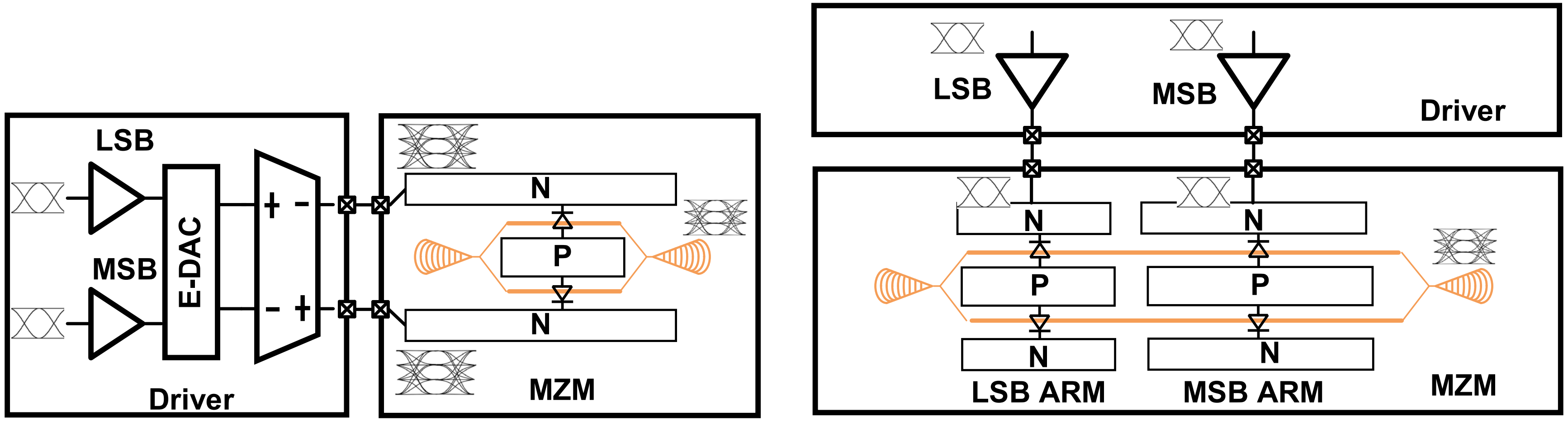
[Cignoli ISSCC 2015]





- Allows for CMOS style drivers
- Well suited for a monolithic silicon photonic process
- Hybrid integration requires may pad connections between CMOS/silicon photonic die

PAM4 Level Generation w/ MZMs

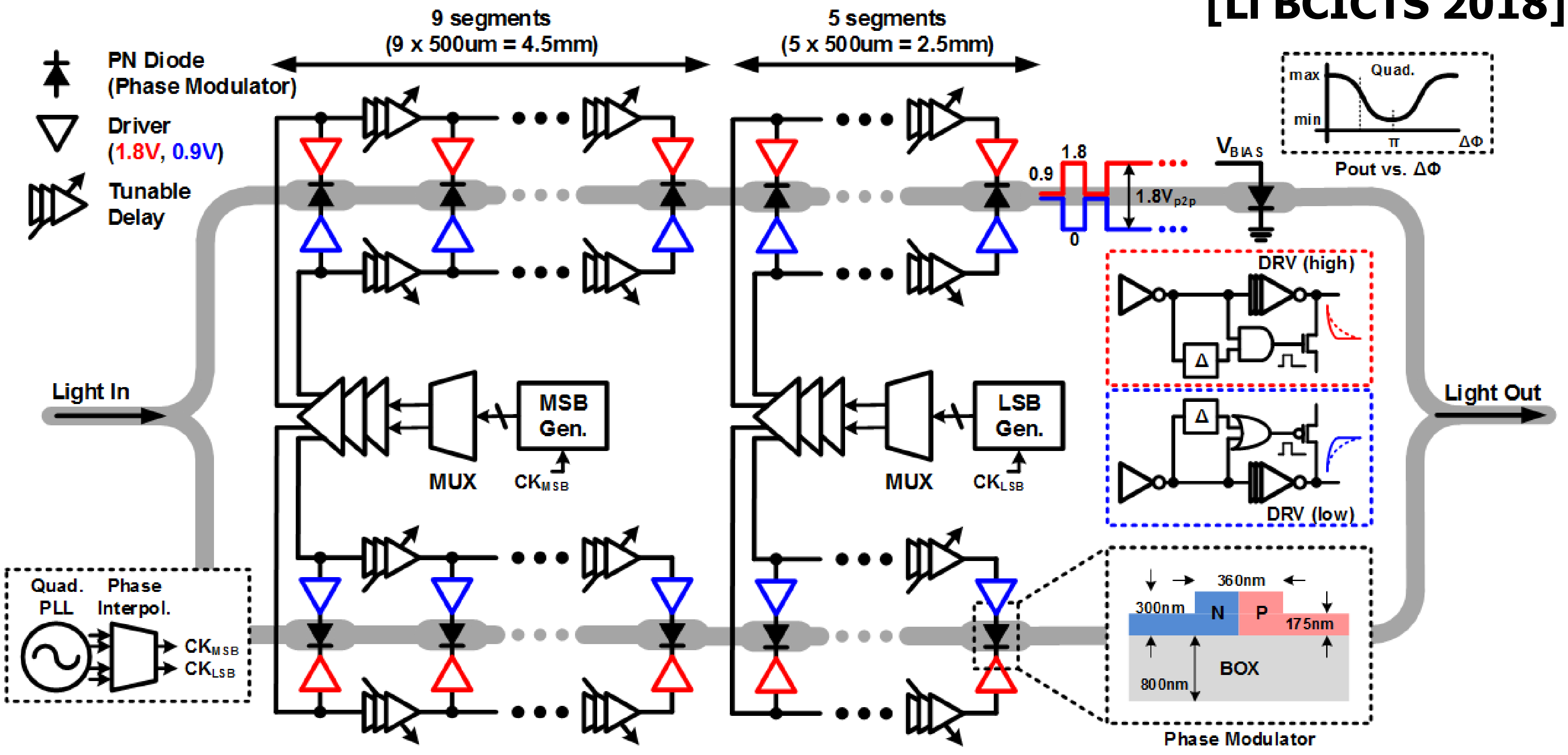


(a) (b)

- E-DAC PAM4 TX
  - PAM4 driver bandwidth and swing limitation
  - Multi current/voltage level
- O-DAC PAM4 TX
  - Velocity mismatch between LSB and MSB
  - Multi driver design

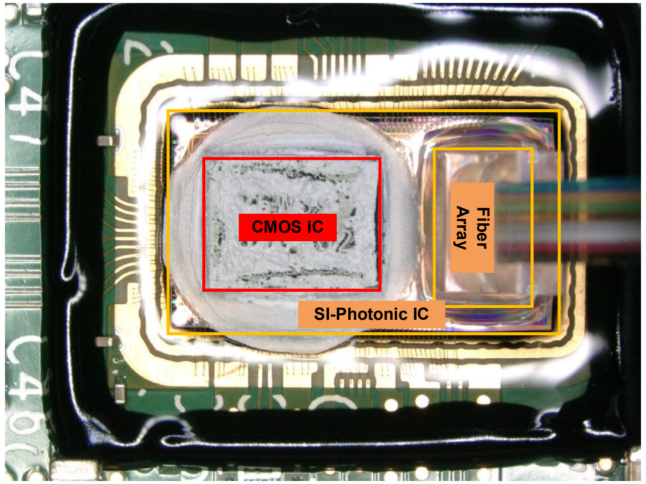
Optical DAC NRZ/PAM4 Reconfigurable MZM TX

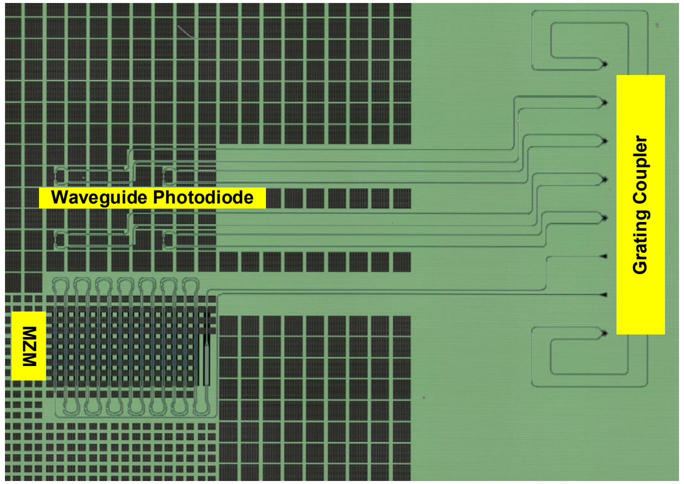
[Li BCICTS 2018]



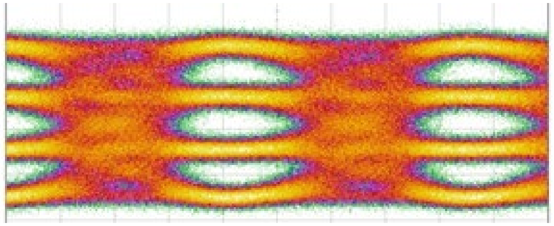
- 5 LSB segments and 9 MSB segments

56Gb/s PAM4 16nm FinFET CMOS Prototype

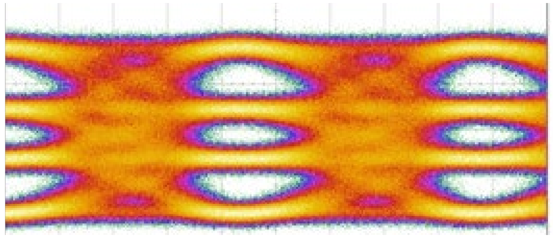




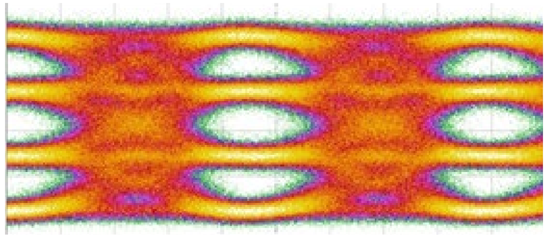
150uw/div 8ps/div



150uw/div 8ps/div

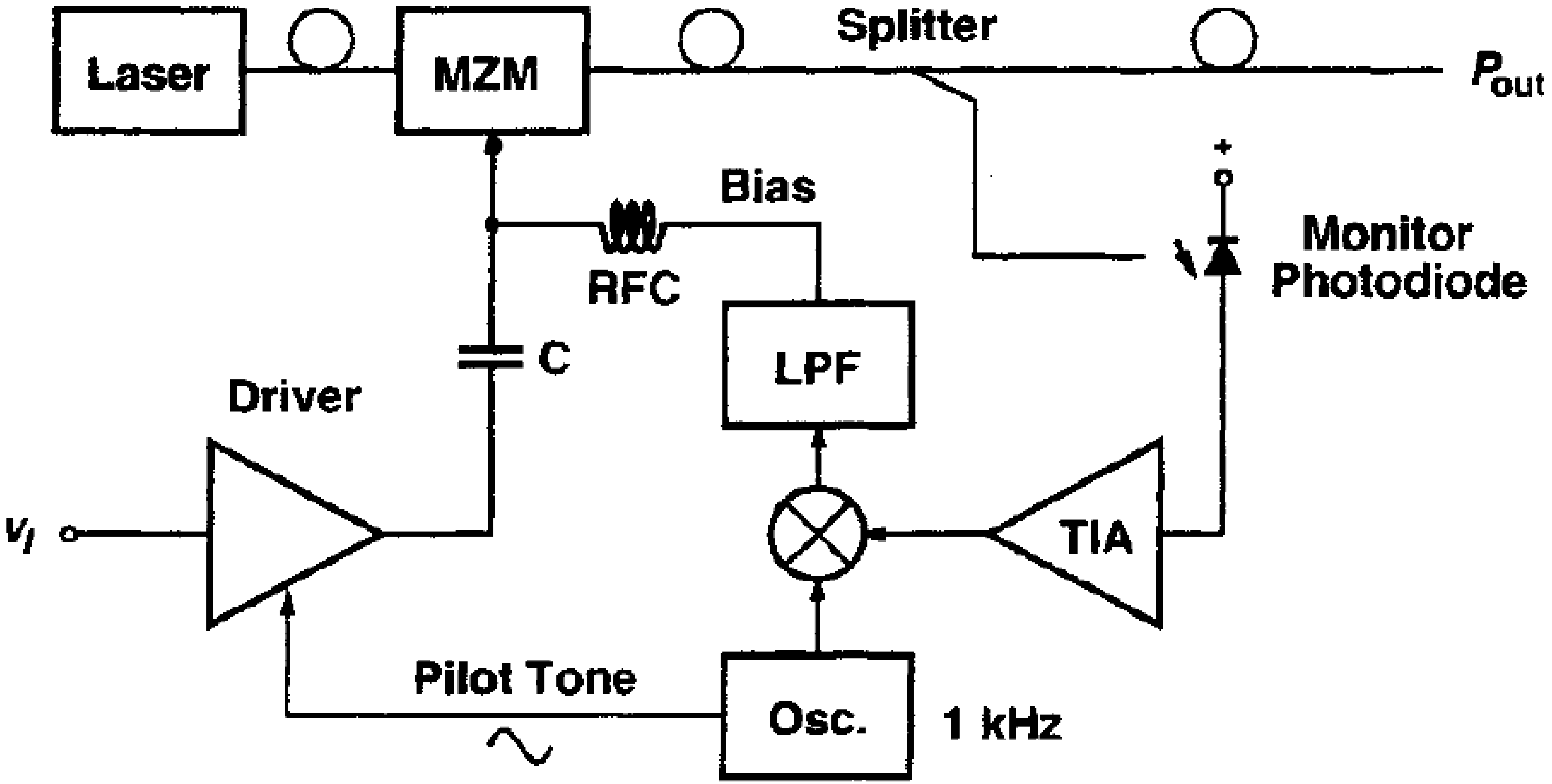


150uw/div 8ps/div



	Segment setting	ER	RLM	EYE width	Eye height
(a)	3 LSB+6 MSB	6.35dB	0.942	5.12ps	11.6uW
(b)	4 LSB+7 MSB	8.14dB	0.896	5.01ps	4.6uW
(c)	4 LSB+8 MSB	8.46dB	0.944	5.7ps	18.4uW

Automatic Bias Control



Next Time

- Electroabsorption Modulator (EAM) TX